IDAT7215: Computer Programming for Product Development and Applications
(Mid Term Exam 13th March 2024)

Student Name: Student ID:

Part I: (50 marks)

Question 1 (30 Marks)

Choose only one of the most appropriate answers for the following questions:

1. Which of the following is a type of program error?

(a) Syntax error. (b) Logical error. (c) Runtime error. (d) All of the above.
Answer: (d) All are possible types of error that can occur while running the program (see first lecture of IDAT7215 for details).

2. Is Python code compiled or interpreted?

(a) Python code is both compiled and interpreted
(b) Python code is neither compiled nor interpreted
(c) Python code is only compiled
(d) Python code is only interpreted
Answer: (a) Many languages have been implemented using both compilers and interpreters, including C, Pascal, and Python.

3. What is a cell in Jupyter Notebook?

(a) A unit of code that can be executed.
(b) A paragraph of narrative text.
(c) A mathematical equation.
(d) A visual representation of data.
Answer: (a) A cell in Jupyter Notebook is a unit of code that can be executed.

4. What is the difference between a Code cell and a Markdown cell in Jupyter Notebook?

(a) Code cells are used for writing code, while Markdown cells are used for writing text.
(b) Code cells are used for writing text, while Markdown cells are used for writing code.
(c) Code cells are used for displaying visualizations, while Markdown cells are used for displaying text.
(d) Code cells are used for writing equations, while Markdown cells are used for displaying data.
Answer: (a) Code cells are used for writing code, while Markdown cells are used for writing text.

5. Which file format does Jupyter Notebook use to store notebooks?
(a) .ipynb (b) .nb (c) .jnb (d) .py
Answer: (a) Jupyter Notebook uses the. ipynb file format to store notebooks.

6. How can you share a Jupyter Notebook with others?

a) By sending them the .ipynb file
b) By exporting the notebook to a different file format, such as HTML or PDF
c) By hosting the notebook on a website or server
d) All of the above

Answer: (d) You can share a Jupyter Notebook with others by sending them the .ipynb file, exporting the notebook to a different file format, such as HTML or PDF, or hosting the notebook on a website or server.

7. Which of the following is not a core data type in Python programming?

(a) Tuples. (b) Class. (c) Lists. (d) Dictionary

Answer: (b) Class is a user-defined data type.

8. What does NumPy stand for in Python?

(a) Numbered Python. (b) Numerical Python. (c) Numbered of python. (d) None

Answer: (b) NumPy stands for Numerical Python and is a Python package for scientific computing.

9. What is the purpose of the NumPy library in Jupyter Notebook?

a) To create visualizations
b) To manipulate and analyze data
c) To perform mathematical calculations
d) To write code

Answer: (c) The NumPy library is used to perform mathematical calculations in Jupyter Notebook.

10. What is the purpose of the panda's library in Jupyter Notebook?

(a) To create visualizations
(b) To manipulate and analyze data
(c) To perform mathematical calculations
(d) To write code

Answer: (b) The panda's library is used to manipulate and analyze data in Jupyter Notebook.

11. Traffic light system is an example of:

(a) Open-loop system (b) Closed-loop system (c) Both (a) and (b) (d) None of these
Answer: (a) The traffic lamp will glow according to the set timing and sequence and is time-dependent. The sequence and time are controlled by relays that work on the pre-programmed time. It does not depend upon the rush of the road. Hence the correct answer is option (a).

12. A control system is stochastic if:

(a) the response to input is unpredictable.
(b) the response to input is predictable and repeatable.
(c) the output varies linearly with the input
(d) the output varies with the time.

Answer: (a) A control system is stochastic if the response to the input is not predictable (see class lecture week3_1_control.pdf for details)

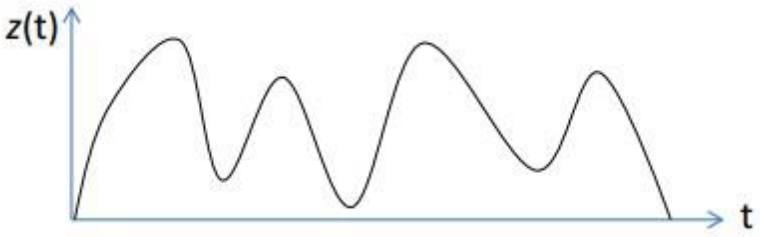


Figure 1: Output of Stochastic Control System

13. In an expert system, the knowledge base component contains the information about the

(a) Rules and Facts (b) Hardware components (c) Software algorithms (d) None
Answer: (a) The knowledge base contains the rules and facts about the knowledge that is gathered from the experts.

14. The key difference between supervised and unsupervised machine learning is

(a) Supervised learning requires labeled data, while unsupervised learning does not.
(d) Supervised learning does not require labeled data, while unsupervised learning does require
(b) Supervised learning predicts labels, while unsupervised learning discovers patterns.
(d) Supervised learning is always more accurate than unsupervised learning.
Answer: (a) The presence or absence of labeled data in the training set distinguishes supervised and unsupervised learning approaches.

15. Which of the following is not a supervised machine learning algorithm?

(a) K-means (b) Naïve Bayes (c) Support Vector Machine (d) Decision Tree
Answer: (a) Decision tree, SVM (Support vector machines) for classification problems and Naïve Bayes are examples of the supervised machine learning algorithm. K-means is an example of an unsupervised machine-learning algorithm.

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Question 2 (20 Marks)

True/False questions.

1. The interpreter translates the entire source code into machine code before execution.
   (a) True (b) False
   Answer: The compiler translates the entire source code into machine code before execution while the interpreter translates code line by line during execution.

2. The interpreter executes faster compared to the compiler.
   (a) True (b) False
   Answer: The compiler translates the entire source code into machine code before execution while the interpreter translates code line by line during execution which makes it slower compared to the compiler.

3. Logical errors are found when the program runs to completion without error but produces the wrong output.
   (a) True (b) False

4. When the order of the denominator polynomial is greater than the numerator polynomial the transfer function is said to be 'proper'.
   (a) True (b) False

5. A control system is deterministic if the response to input is predictable and repeatable.
   (a) True (b) False

6. If all the poles of the system lie in the left half plane the system is said to be unstable.
   (a) True (b) False
   Answer: If all the poles of the system lie in the left half plane the system is said to be stable.

7. Expert systems are only as good as the data and rules they are given.
   (a) True (b) False

8. The main purpose of the Interference Engine in an expert system is to input data and interact with the user.
   (a) True (b) False
   Answer: The main purpose of the User Interface in an expert system is to input data and interact with the user.

9. Support Vector Machine (SVM) and Random Forest algorithms are examples of supervised machine learning.
   (a) True (b) False

10. Recommendations systems are a classic example of unsupervised machine learning.
    (a) True (b) False

Part II: (50 marks)

Question 3 (30 marks):

Answer the following questions about Python Programming.

3.1 What are the differences between lists and sets in Python? **(6 Marks)**

Answer:

List: A list contains an arbitrary number of elements that are stored in sequential order. The elements don't need to be of the same type. The list is mutable.

Sets: The set represents a collection of unique elements. The set can be created by listing its elements between curly brackets. Sets are mutable.

Type	Example	Description	Can we change the content within this collection?	Are the items in the collection ordered stored?
list	[1, 2, 3]	List: an ***mutable***, ***ordered*** collection	Yes – mutable	Yes – ordered
tuple	(1, 2, 3)	Tuple: an ***immutable, ordered*** collection	No – immutable	Yes – ordered
set	{1, 2, 3}	Set: ***Unordered*** collection of unique values	Yes – mutable	No – unordered
dict	{'a':1, 'b':2, 'c':3}	Dictionary: ***Unordered*** (key, value) mapping	Yes – mutable	No – unordered

3.2 What will be the output of the following Python code? **(4 marks)**

```
a = [1, 14, 3, 16, 18, 4, 19, 17, 7, 6, 11, 15, 8, 10, 2]
a.sort(reverse=True)
print(a[9:3:-3])
```

Ans: [7,11]

3.3 What will be the output of the following Python code? **(4 marks)**

```
import numpy as np
a=np.array([[[1,2,3], [4,5,6]], [[1,2,3], [4,5,6]]])
print(a.ndim)
print(a.shape)
print(a.size)
```

Ans: print(a.ndim)=3, print(a.shape)=(2,2,3), print(a.size)=12

3.4 What will be the output of the following Python code? **(4 marks)**

```
import numpy as np
a = np.arange(25).reshape(5,5) # Define 5*5 matrix using Numpy Library
row = np.array([2,4])
col = np.array([0,3])
print(a[row,col])
```

Ans: [10 23]

3.5 What will be the output of the following Python code? **(4 marks)**

```
a = [1, 6, 3, 9, 7, 8]
a.sort(reverse=True)
print(a[5:2:-2])
```

Ans: [1, 6]

3.6 What will be the output of the following Python code? **(4 marks)**

```
def my_print(person = "Joe"):
    message = "Hello, " + person
    print(message)

my_print("Peter")
```

Ans: Hello, Peter

3.7 What will be the output of the following Python code? **(4 marks)**

```
import pandas as pd
df = pd.DataFrame([[1000, "Jack", 21], [1500, "John", 29]], columns=["1", "2", "3"])
df
```

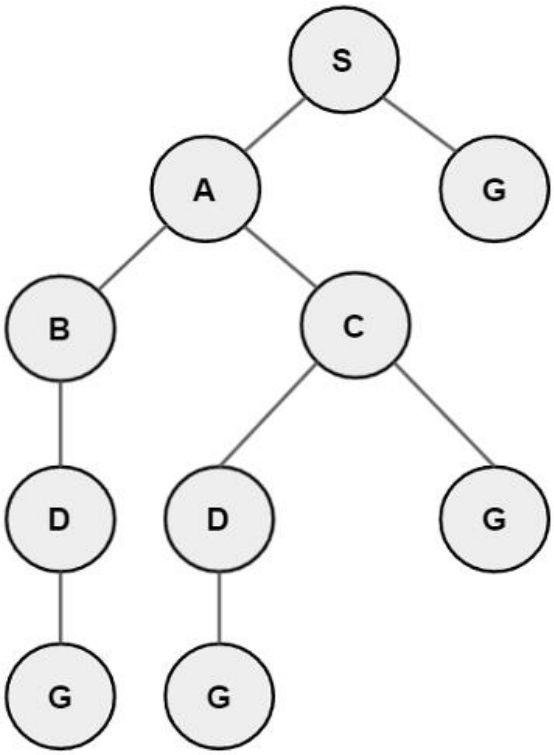
	1	2	3
0	1000	Jack	21
1	1500	John	29

```
print(df.iloc[1,1])
```

Ans: John

Question 4 (10 marks)

Look at the graph below and traverse nodes in the tree starting from node "S" using:
1) **breadth-first search**; 2) **depth-first search**.



ANSWER

1) breadth-first search: S,A,G, B,C,D,D,G,G,G
2) depth-first search: S,A,B,D,G, C,D,G,G,G

Alternative answers are also acceptable:
1) breadth-first search: S,A,G
2) depth-first search: S,A,B,D,G

Question-5 (10 marks)

Suppose you have the following rules in your knowledge base, and we are using forward chaining (data-driven) what "goal" would the system return if A, B, C, and E are true? Example of how the results are obtained (e.g., show the reasoning path). (5 marks)

Rule 1: IF X is true
AND B is true
AND E is true
THEN Y is true

Rule 2: IF Y is true
AND D is true
THEN Z is true

Rule 3: IF A is true
THEN X is true

Rule 4: IF Y is true
AND C is true
THEN W is true

Answer:
The system will return "W" as the goal. Since A is true then by following Rule 3 X is true. Now if X, B, and E are true then following Rule 1 Y is true. Now If Y and C are True then following Rule 4 W is true. Rule 2 is not applicable as we do not know the status of D. So, the reasoning path is.

Rule 3 $\Longrightarrow$ Rule 1 $\Longrightarrow$ Rule 4.

END OF EXAM-